



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

April 2012

Assessment Task 2
Year 12

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators may be used.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question if full marks are to be awarded.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 60

- Attempt sections A – C.
- Start each **NEW** section in a separate answer booklet.
- Hand in your answers in 3 separate bundles:

Section A
Section B
Section C

Examiner: *J. Chen*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

START A NEW ANSWER BOOKLET

SECTION A [20 marks]

Marks

For these 10 questions there is one correct answer per question. Write down in your answer booklet the question number and letter of your answer.

1.

[1]

$$\int_1^2 \frac{dx}{2x+5}$$

equals

(a) $\ln\left(\frac{9}{7}\right)$

(b) $\frac{1}{2}\ln(63)$

(c) $\frac{1}{2}\ln\left(\frac{9}{7}\right)$

(d) $\ln(63)$

2.

[1]

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$$

equals

(a) 2

(b) 1

(c) 0

(d) $\frac{1}{2}$

3. If $\log_m 64 + \log_m 4 = x \log_m 2$, then the value of x is:

[1]

(a) 4

(b) 8

(c) 6

(d) 2

4. $\frac{d}{dx} \log_e(e^{3x} + 2)$ equals [1]

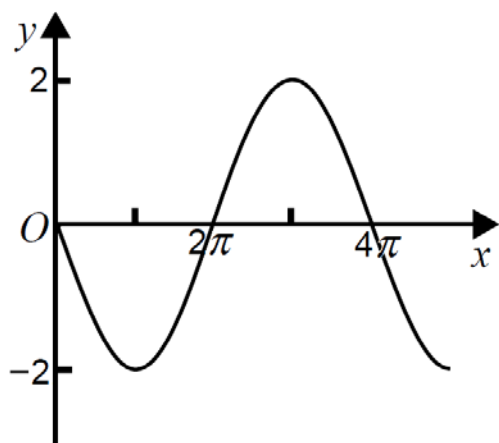
(a) $3e^{3x}$

(b) $e^{3x} + 2$

(c) $\frac{1}{e^{3x}+2}$

(d) $\frac{3e^{3x}}{e^{3x}+2}$

5. The diagram below shows a part of the graph of a trigonometric function. [1]



A possible equation for the function is

(a) $y = 2 \sin 2x$

(b) $y = -2 \cos 2x$

(c) $y = -2 \sin \frac{x}{2}$

(d) $y = 2 \cos \frac{x}{2}$

6. [1]

$$\int \cos 6x \cdot dx$$

equals

(a) $\frac{\sin 6x}{6} + C$

(b) $-\frac{\sin 6x}{6} + C$

(c) $6 \sin 6x + C$

(d) $-6 \sin 6x + C$

7.

[1]

$$\int 8xe^{x^2} \cdot dx$$

equals

(a) $4xe^{x^2} + C$

(b) $8e^{x^2} + C$

(c) $2xe^{x^2} + C$

(d) None of the above

8. What is the exact value of $\sin 75^\circ$?

[1]

(a) $\frac{\sqrt{2}+\sqrt{6}}{4}$

(b) $\frac{\sqrt{2}-\sqrt{6}}{4}$

(c) $\frac{\sqrt{6}+\sqrt{2}}{4}$

(d) $\frac{\sqrt{6}-\sqrt{2}}{4}$

9.

[1]

$$\int_{-\pi}^{\pi} 2 \sin x \cdot dx$$

equals

(a) 0

(b) 2

(c) $2 \int_0^{\pi} 2 \sin x \cdot dx$

(d) $\left| \int_{-\pi}^0 2 \sin x \cdot dx \right| + \int_0^{\pi} 2 \sin x \cdot dx$

10. If $f(x) = \cos 2x$, then $f' \left(-\frac{\pi}{6} \right)$ is:

[1]

(a) $\frac{\sqrt{3}}{2}$

(b) $\sqrt{3}$

(c) $-\frac{\sqrt{3}}{2}$

(d) None of the above

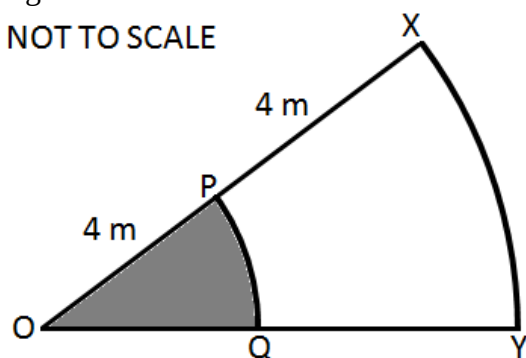
End of Multiple Choice Section

11. Differentiate $\cot x$. [2]

12. Solve the equation, [3]
$$3 \ln(x + 1) = \ln(x^3 + 19)$$

13. Find the equation of the tangent to the curve $y = \sin x$ at $x = \pi$. [2]

14. PQ and XY are arcs of concentric circles with centre O. $OP = PX = 4$ m. [3]
The shaded sector OPQ has area $\frac{2\pi}{3}$ square metres. Find $\angle POQ$ in degrees.



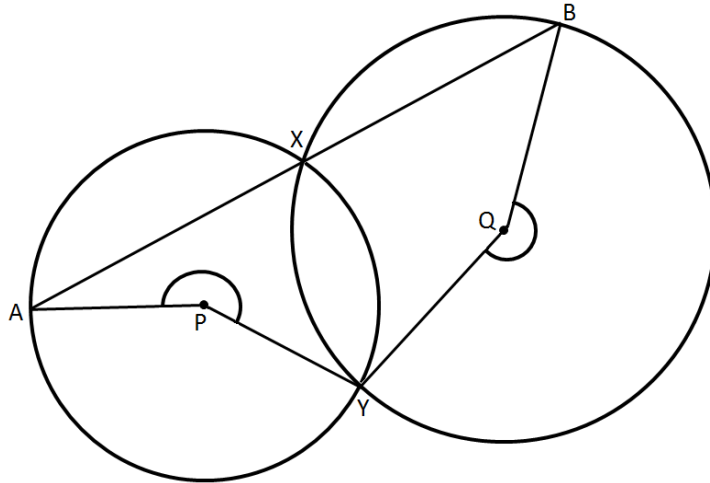
End of Section A

START A NEW ANSWER BOOKLET

SECTION B [20 marks]

Marks

- At any point on the curve $y = f(x)$ the gradient function is given by $\frac{dy}{dx} = \frac{x+1}{x+2}$. If $y = -1$ when $x = -1$, find the value of y when $x = 1$, correct your answer to the nearest 3 significant figures. [4]
- P and Q are centres of the circles, AXB is a straight line. Prove that $\angle APY = \angle BQY$ as marked below. [3]



- Evaluate [2]

$$\int_0^{\frac{\pi}{6}} \sec^2 x \tan^8 x \cdot dx$$

- Consider the function $f(x) = \frac{\log_e x}{x^2}$. [6]
 - Find the x intercept of the curve.
 - Find the coordinates of the turning point and the point of inflexion.
 - Hence, sketch the curve $y = f(x)$ and label the critical points and any asymptotes.
- Consider the function $f(x) = x - \sin x$. [2]
 $P(X, 1)$ is a point on the curve $y = f(x)$. Starting with an initial approximation of $X = 2$, use one application of Newton's Method to find an improved approximation to the value of X , giving the answer correct to 3 decimal places.
- Prove by Mathematical Induction that $3^{3n} + 2^{n+2}$ is divisible by 5 for all integers $n \geq 1$. [3]

End of Section B

START A NEW ANSWER BOOKLET

SECTION C [20 marks]

Marks
[2]

1.

- (i) Show that there is a solution to the equation $x - 2 = \sin x$ between $x = 2.5$ and $x = 2.6$.
- (ii) By halving the interval, find the solution correct to 2 decimal places.

2.

[5]

- (i) Use the Principle of Mathematical Induction to prove that
$$\sin(x + n\pi) = (-1)^n \sin x$$
for all positive integers n .

- (ii) If

$$S = \sum_{k=1}^n \sin(x + k\pi)$$

for $0 < x < \frac{\pi}{2}$ and for all positive integers n .
Prove that $-1 < S \leq 0$.

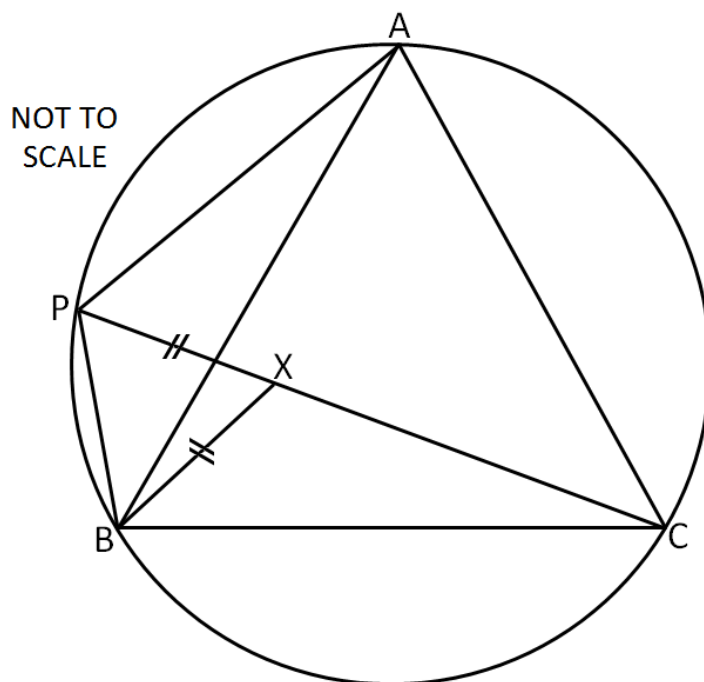
3. Consider the function $f(x) = e^x \left(1 - \frac{x}{4}\right)^4$.

[8]

- (i) Find the coordinates of the stationary points and determine their nature.
- (ii) Sketch the curve $y = f(x)$ and label the turning points and any asymptotes.
- (iii) Hence, prove that $\left(\frac{5}{4}\right)^4 \leq e \leq \left(\frac{4}{3}\right)^4$.

4. In the diagram, A, B, C and P are points on the circumference of the circle and $\triangle ABC$ is an equilateral triangle. X is a point on the straight line PC such that $PX = BX$. Prove that $PC = PA + PB$.

[5]



Copy or trace the diagram into your answer booklet.

End of Section C
End of Exam

SECTION A

$$\begin{aligned}
 1. \quad \int_1^2 \frac{dx}{2x+5} &= \frac{1}{2} \left[\ln(2x+5) \right]_1^2 \\
 &= \frac{1}{2} \{ \ln 9 - \ln 7 \} \\
 &= \frac{1}{2} \ln \frac{9}{7} \quad \text{(C)}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad \lim_{x \rightarrow 0} \frac{\sin 2x}{x} &= 2 \lim_{2x \rightarrow 0} \frac{\sin 2x}{2x} \\
 &= 2 \times 1 \\
 &= 2 \quad \text{(A)}
 \end{aligned}$$

$$\begin{aligned}
 3. \quad \log_m 64 + \log_m 4 &= x \log_m 2 \\
 \text{LHS} &= \log_m 256 \\
 &= \log_m 2^8 \\
 &= 8 \log_m 2 \\
 x &= 8 \quad \text{(B)}
 \end{aligned}$$

$$\begin{aligned}
 4. \quad \frac{d}{dx} \log_e (e^{3x} + 2) \\
 &= \frac{1}{e^{3x} + 2} \cdot e^{3x} \cdot 3 \\
 &= \frac{3e^{3x}}{e^{3x} + 2} \quad \text{(D)}
 \end{aligned}$$

$$5. \quad \text{C} \quad y = -2 \sin \frac{x}{2}$$

$$6. \int \cos 6x \, dx$$

$$= \frac{1}{6} \sin 6x + C \quad \textcircled{A}$$

$$7. \int 8x e^{x^2} \, dx$$

$$\text{Let } u = x^2$$

$$du = 2x \, dx$$

$$= 4 \int e^u \, du$$

$$= 4e^u + C$$

$$= 4e^{x^2} + C$$

$\textcircled{D}$. None of the above

$$\textcircled{8} \quad \sin 75^\circ = \sin (45 + 30)^\circ$$

$$= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$$

$$= \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

$\textcircled{A}$ or $\textcircled{C}$

$$9. \int_{-\pi}^{\pi} 2 \sin x \, dx$$

$$= 2 \left[-\cos x \right]_{-\pi}^{\pi}$$

$$= 2 \left[-\cos \pi - (-\cos(-\pi)) \right]$$

$$= 2 \left[1 + -1 \right]$$

$$= 0 \quad \textcircled{A}$$

$$10 \quad f(x) = \cos 2x$$

$$f'(x) = -\sin 2x \cdot 2$$

$$f'(-\frac{\pi}{6}) = -\sin(-\frac{\pi}{3}) \cdot 2$$

$$= -2 \times -\sin \frac{\pi}{3}$$

$$= 2 \times \frac{\sqrt{3}}{2}$$

$$= \sqrt{3}$$

(B)

$$11. \quad \frac{d}{dx} (\cot x) = \frac{d}{dx} \left(\frac{\cos x}{\sin x} \right)$$

$$= \frac{\sin x \cdot -\sin x - \cos x \cdot \cos x}{\sin^2 x}$$

$$= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x}$$

$$= \frac{-1}{\sin^2 x}$$

$$= -\operatorname{cosec}^2 x.$$

(2)

$$12. \quad 3 \ln(x+1) = \ln(x^3+19)$$

$$\therefore \ln((x+1)^3) = \ln(x^3+19)$$

$$\therefore x^3 + 3x^2 + 3x + 1 = x^3 + 19$$

$$\therefore 3x^2 + 3x - 18 = 0$$

$$x^2 + x - 6 = 0$$

$$(x+3)(x-2) = 0$$

$$x = -3 \text{ or } x = 2$$

$$\text{As } x+1 > 0, \quad x \neq -3$$

$$\therefore \text{Soln : } x = 2$$

(3)

$$13. \quad y = \sin x$$

$$y' = \cos x$$

$$\text{When } x = \pi, \quad y = \sin \pi = 0$$

$$y' = \cos \pi = -1$$

$$\therefore \text{Eqn of tangent is } y - 0 = -1(x - \pi)$$

$$y = -x + \pi$$

$$y + x = \pi$$

2

$$14. \quad \text{Area} = \frac{1}{2} r^2 \theta$$

$$= \frac{1}{2} \times 16 \times \theta = \frac{2\pi}{3}$$

$$\therefore \theta = \frac{2\pi}{3}$$

$$\therefore \theta = \frac{2\pi}{24}$$

$$= \frac{\pi}{12}$$

$$= 15^\circ$$

$$\therefore \angle POQ = 15^\circ$$

3

2012 Extension 1 Mathematics Task 2:
Solutions— Section B

1. At any point on the curve $y = f(x)$ the gradient function is given by $\frac{dy}{dx} = \frac{x+1}{x+2}$. If $y = -1$ when $x = -1$, find the value of y when $x = 1$, correct your answer to the nearest 3 significant figures.

4

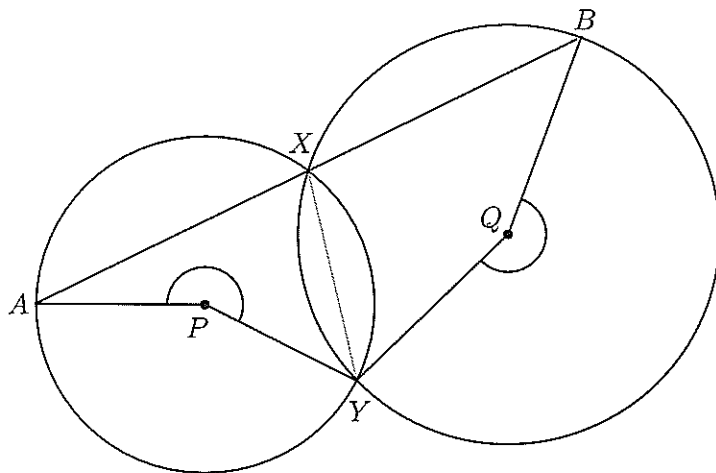
Solution:

$$\begin{aligned}\frac{dy}{dx} &= 1 - \frac{1}{x+2}, \\ y &= x - \ln(x+2) + c, \\ -1 &= -1 - \ln 1 + c, \\ c &= 0, \\ y &= x - \ln(x+2), \\ &= 1 - \ln 3 \text{ when } x = 1, \\ &\approx -0.0986 \text{ (3 sig. fig.)}\end{aligned}$$

$$-2 \left| \begin{array}{cc} 1 & 1 \\ & -2 \\ \hline & 1 & -1 \end{array} \right|$$

2. P and Q are centres of the circles, AXB is a straight line. Prove that $\angle APY = \angle BQY$ as marked below.

3



Solution:

$$\begin{aligned}\widehat{APY} &= 2\widehat{AXY} \text{ (}\angle \text{ at centre } 2 \times \angle \text{ at circumf.)}, \\ 360^\circ - \widehat{BQY} &= 2\widehat{BXY} \text{ (}\angle \text{ at centre } 2 \times \angle \text{ at circumf.)}, \\ \widehat{AXY} + \widehat{BXY} &= 180^\circ \text{ (}\widehat{AXB} \text{ is straight)}, \\ \widehat{APY} + 360^\circ - \widehat{BQY} &= 2 \times 180^\circ, \\ \widehat{APY} &= \widehat{BQY}, \\ \therefore \text{reflex } \widehat{APY} &= \text{reflex } \widehat{BQY}.\end{aligned}$$

3. Evaluate $\int_0^{\pi/6} \sec^2 x \tan^8 x \, dx$

2

Solution: $I = \int_0^{\frac{1}{\sqrt{3}}} u^8 \cdot du,$

$$= \left[\frac{u^9}{9} \right]_0^{\frac{1}{\sqrt{3}}},$$

$$= \frac{1}{9} \times \frac{1}{81\sqrt{3}} - 0,$$

$$= \frac{\sqrt{3}}{2187}.$$

put $u = \tan x$
 $du/dx = \sec^2 x$
 when $x = \pi/6, \quad u = 1/\sqrt{3}$
 $x = 0, \quad u = 0$

4. Consider the function $f(x) = \frac{\log_e x}{x^2}.$

6

(a) Find the x intercept of the curve.

Solution: $\ln x = 0$ when $x = 1$, so the x -intercept is at $(1, 0)$.

(b) Find the coordinates of the turning point and the point of inflexion.

Solution: $f'(x) = \frac{\frac{x^2}{x} - 2x \ln x}{x^4}, \quad f''(x) = \frac{x^3 \left(\frac{-2}{x} \right) - 3x^2(1 - 2 \ln x)}{x^6},$

$$= \frac{1 - 2 \ln x}{x^3}, \quad = \frac{-2 - 3 + 6 \ln x}{x^4},$$

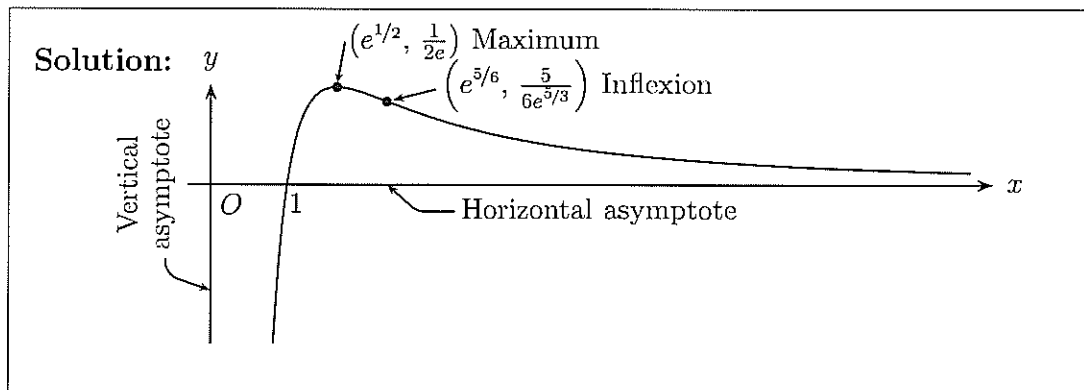
$$= 0 \text{ when } x = e^{1/2}. \quad = \frac{6 \ln x - 5}{x^4},$$

$$= 0 \text{ when } x = e^{5/6}.$$

$\therefore$ Maximum $\left(e^{1/2}, \frac{1}{2e} \right), \quad f''(e^{1/2}) = \frac{-2}{e^2} < 0.$

Inflexion $\left(e^{5/6}, \frac{5}{6e^{5/3}} \right),$

(c) Hence sketch the curve $y = f(x)$, and label the critical points and any asymptotes.



5. Consider the function $f(x) = x - \sin x$.

2

$P(X, 1)$ is a point on the curve $y = f(x)$. Starting with an initial approximation of $X = 2$, use one application of Newton's Method to find an improved approximation to the value of X , giving the answer correct to 3 decimal places.

Solution: $f'(x) = 1 - \cos x$.

$$a_1 = 2 - \frac{2 - \sin 2 - 1}{1 - \cos 2},$$

$$\approx 1.936 \text{ (3 dec. pl.)}$$

6. Prove by Mathematical Induction that $3^{3n} + 2^{n+2}$ is divisible by 5 for all integers $n \geq 1$.

3

Solution: $S_n = 3^{3n} + 2^{n+2}$.

$$\begin{aligned} \text{Test } n = 1, \quad S_1 &= 3^3 + 2^3, \\ &= 27 + 8, \\ &= 35. \end{aligned}$$

$\therefore$ True for $n = 1$.

Assume true for $n = k$,

$$\text{i.e. } S_k = 5p \text{ where } p \in \mathbb{Z}.$$

$$\text{Test } n = k + 1,$$

$$\text{i.e. } S_{k+1} = 5q \text{ where } q \in \mathbb{Z}.$$

$$\begin{aligned} \text{L.H.S.} &= 3^{3(k+1)} + 2^{k+1+2}, \\ &= 3^{3k+3} + 2^{k+3}, \\ &= 27 \cdot 3^{3k} + 2 \cdot 2^{k+2}, \\ &= 27 (3^{3k} + 2^{k+2}) - 25 \cdot 2^{k+2}, \\ &= 27 S_k - 25 \cdot 2^{k+2}, \\ &= 27 \cdot 5p - 25 \cdot 2^{k+2} \text{ (using the assumption),} \\ &= 5 (27p - 5 \cdot 2^{k+2}), \\ &= 5q. \end{aligned}$$

So, true for $n = k + 1$ if true for $n = k$; true for $n = 1$,
and so true for $n = 2, 3, \dots$, for all $n \geq 1$.

Section C Solutions

1.

- (i) Show that there is a solution to the equation $x - 2 = \sin x$ between $x = 2.5$ and $x = 2.6$.
- (ii) By halving the interval, find the solution correct to 2 decimal places.

$$\text{Let } f(x) = x - 2 - \sin x$$

$$f(2.5) = -0.098472144$$

$$f(2.6) = 0.084498628$$

As $f(x)$ is continuous and $f(2.5).f(2.6) < 0$ then there is a solution for $2.5 < x < 2.6$

One application of the “halving the interval” gives an approximation as $x = 2.55$.
[This is probably what the question meant for 1 mark]

However, what the question is really asking means that we have to find the correct solution rounded to 2 d.p.

Is $x = 2.55$ the correct solution rounded to 2 dp?

$$f(2.55) = -0.007683717$$

So a smaller interval containing the solution is $2.55 < x < 2.6$ and so a second approximation would be $x = 2.575$

As $f(2.575) = 0.038239727$ then a smaller subinterval containing the solution is $2.55 < x < 2.575$.

Using the table below, the correct solution to 2 dp is 2.55

a	b	$f(a)$	$f(b)$	$f(a) \times f(b)$	midpoint	$f(\text{midpoint})$
2.5	2.6	-0.098472144	0.084498628	–	2.55	–
2.55	2.6	-0.007683717	8.863738035	–	2.575	+
2.55	2.575	-0.007683717	8.556317158	–	2.5625	+
2.55	2.5625	-0.007683717	8.405697317	–	2.55625	+
2.55	2.55625	-0.007683717	8.331148842	–	2.553125	–
2.553125	2.55625	-0.001962081	8.331148842	–	2.5546875	+
2.553125	2.5546875	-0.001962081	8.312590505	–	2.55390625	–

2.

- (i) Use the Principle of Mathematical Induction to prove that
 $\sin(x + n\pi) = (-1)^n \sin x$
 for all positive integers n .

Test $n = 1$

$$\text{LHS} = \sin(x + \pi) = -\sin x \quad (3^{\text{rd}} \text{ quadrant results})$$

$$\text{RHS} = (-1)^1 \sin x = -\sin x$$

$\therefore$ true for $n = 1$

Assume true for $n = k$ i.e. $\sin(x + k\pi) = (-1)^k \sin x$

Need to prove true for $n = k + 1$ i.e. $\sin[x + (k + 1)\pi] = (-1)^{k+1} \sin x$

$$\begin{aligned} \text{LHS} &= \sin[x + (k + 1)\pi] \\ &= \sin[(x + k\pi) + \pi] \\ &= -\sin(x + k\pi) \\ &= -(-1)^k \sin x \quad [\text{from assumption}] \\ &= (-1)^{k+1} \sin x \\ &= \text{RHS} \end{aligned}$$

So the formula is true for $n = k + 1$ when it is true for $n = k$.

By the principle of mathematical induction the formula is true for all positive integers.

- (ii) If

$$S = \sum_{k=1}^n \sin(x + k\pi)$$

for $0 < x < \frac{\pi}{2}$ and for all positive integers n .

Prove that $-1 < S \leq 0$.

For $0 < x < \frac{\pi}{2}$, $\sin x > 0$, but $\sin x \neq 1$

$$\begin{aligned} S &= \sum_{k=1}^n \sin(x + k\pi) \\ &= \sum_{k=1}^n (-1)^k \sin x \quad [\text{From (i)}] \\ &= \sin x \times \sum_{k=1}^n (-1)^k \end{aligned}$$

If n is even then $\sum_{k=1}^n (-1)^k = 0$ and if n is odd then $\sum_{k=1}^n (-1)^k = -1$

$$\therefore -1 < \sin x \sum_{k=1}^n (-1)^k \leq 0 \quad [\text{As indicated } \sin x \neq 1, \text{ so } S \neq -1]$$

$$\therefore -1 < S \leq 0$$

3. Consider the function $f(x) = e^x \left(1 - \frac{x}{4}\right)^4$.

- (i) Find the coordinates of the stationary points and determine their nature.

$$f(x) = e^x \left(1 - \frac{x}{4}\right)^4$$

$$\begin{aligned} f'(x) &= e^x \times 4 \left(1 - \frac{x}{4}\right)^3 \times \left(-\frac{1}{4}\right) + e^x \left(1 - \frac{x}{4}\right)^4 \\ &= -e^x \left(1 - \frac{x}{4}\right)^3 + e^x \left(1 - \frac{x}{4}\right)^4 \\ &= e^x \left(1 - \frac{x}{4}\right)^3 \left[\left(1 - \frac{x}{4}\right) - 1\right] \\ &= -\frac{x}{4} e^x \left(1 - \frac{x}{4}\right)^3 \end{aligned}$$

Stationary points occur when $f'(x) = 0$ i.e. $-\frac{x}{4} e^x \left(1 - \frac{x}{4}\right)^3 = 0$

$$\therefore x = 0, 4$$

NB $e^x > 0$ for all x , so this has been ignored from the calculations.

x	-1	0	1	3	4	5
y'	$\frac{1}{4} \left(\frac{5}{4}\right)^3$	0	$-\frac{1}{4} \left(\frac{3}{4}\right)^3$	$-\frac{3}{4} \left(\frac{1}{4}\right)^3$	0	$\frac{5}{4} \left(\frac{1}{4}\right)^3$
	+	0	-	-		+

$$f(0) = 1$$

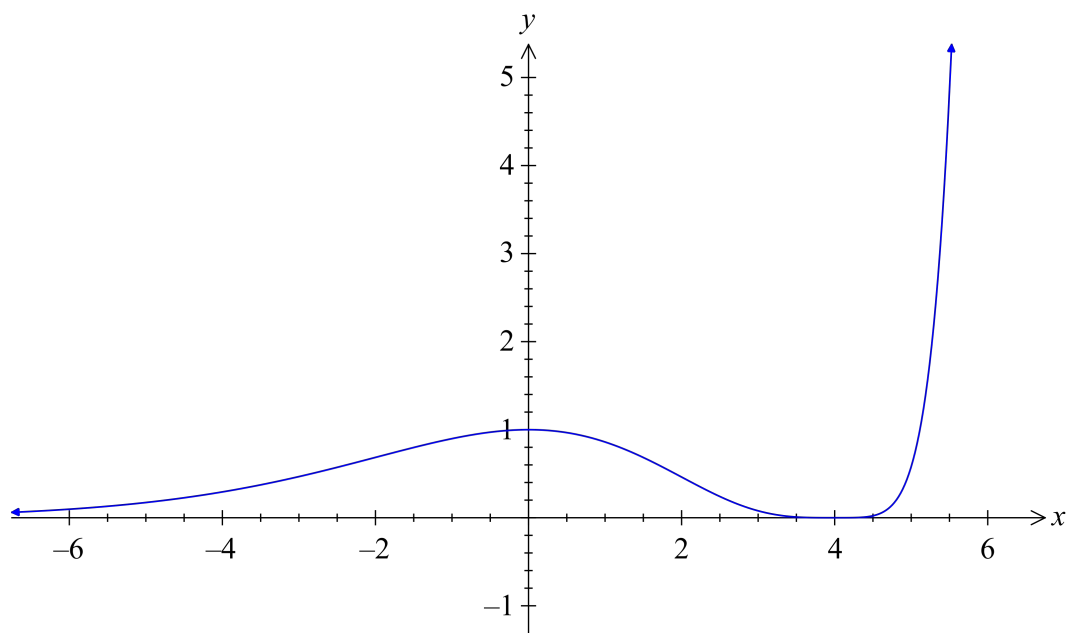
$$f(4) = 0$$

$\therefore (0, 1)$ is a maximum turning point and $(4, 0)$ is a minimum turning point.

- (ii) Sketch the curve $y = f(x)$ and label the turning points and any asymptotes.

As $x \rightarrow -\infty$, $f(x) \rightarrow 0^+$, and as $x \rightarrow \infty$, $f(x) \rightarrow \infty$

So the horizontal asymptote is $y = 0$.



(iii) Hence, prove that $\left(\frac{5}{4}\right)^4 \leq e \leq \left(\frac{4}{3}\right)^4$.

$$f(x) = e^x \left(1 - \frac{x}{4}\right)^4$$

$$f'(x) = e^x \times 4 \left(1 - \frac{x}{4}\right)^3 \times \left(-\frac{1}{4}\right) + e^x \left(1 - \frac{x}{4}\right)^4$$

$$= -e^x \left(1 - \frac{x}{4}\right)^3 + e^x \left(1 - \frac{x}{4}\right)^4$$

$$= e^x \left(1 - \frac{x}{4}\right)^3 \left[\left(1 - \frac{x}{4}\right) - 1\right]$$

$$= -\frac{x}{4} e^x \left(1 - \frac{x}{4}\right)^3$$

So from the graph, $f(-1) \leq 1$ i.e. $e^{-1} \left(1 + \frac{1}{4}\right)^4 \leq 1$

$$\therefore \left(\frac{5}{4}\right)^4 \leq e$$

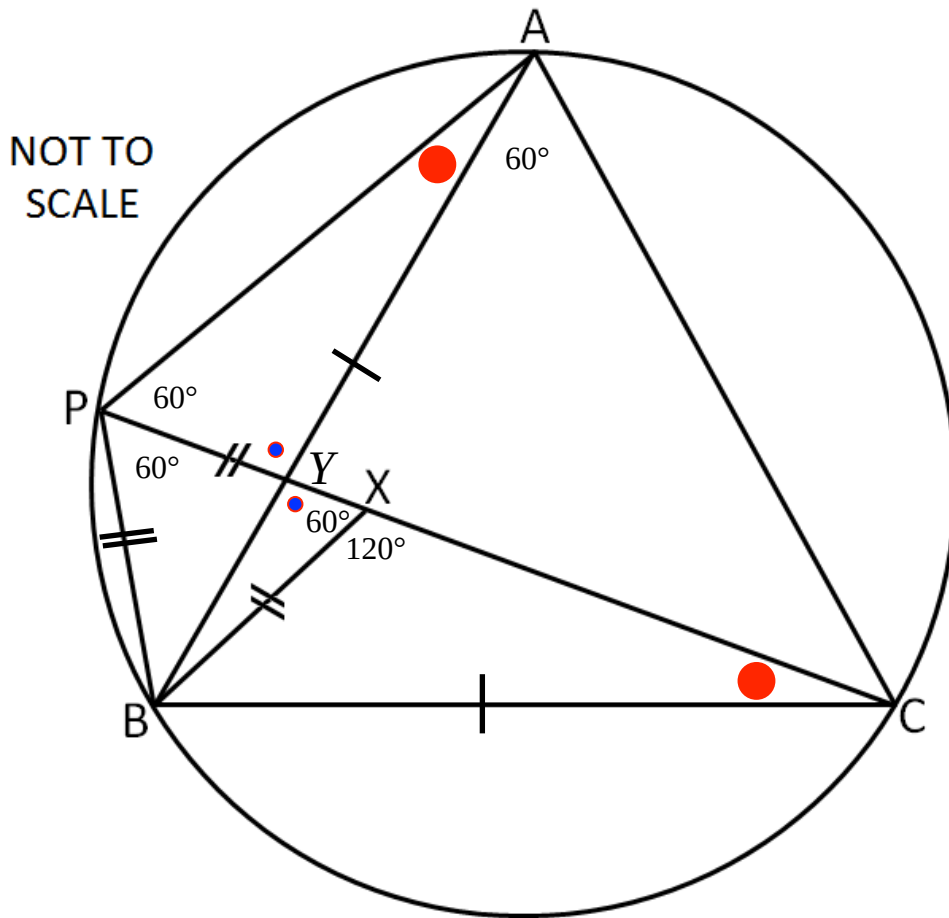
Also from the graph $f(1) \leq 1$ i.e. $e \left(\frac{3}{4}\right)^4 \leq 1$

$$\therefore e \leq \left(\frac{4}{3}\right)^4$$

$$\therefore e \leq \left(\frac{4}{3}\right)^4$$

$$\therefore \left(\frac{5}{4}\right)^4 \leq e \leq \left(\frac{4}{3}\right)^4$$

4. In the diagram, A, B, C and P are points on the circumference of the circle and $\triangle ABC$ is an equilateral triangle. X is a point on the straight line PC such that $PX = BX$. Prove that $PC = PA + PB$.



As $\triangle ABC$ is equilateral then $\angle BAC = 60^\circ$

$$\therefore \angle BPX = 60^\circ$$

(angles in the same segment)

Similarly, $\angle APC = \angle ABC = 60^\circ$

$PX = PB$ means that $\angle PBX = 60^\circ$

(equal angles opposite equal sides)

$$\therefore \angle PXB = 60^\circ$$

($\angle$ sum $\triangle PXB$)

$\therefore \triangle PXB$ is equilateral and $PX = PB = XB$

Now $\angle PAB = \angle PCB$

(angles in the same segment)

$$\angle APB = \angle BPX + \angle APC = 120^\circ$$

(adjacent angles)

$$\angle BXC = 120^\circ$$

(angle sum straight angle, $\angle PXC$)

In $\triangle PAB$ and $\triangle BXC$

$$PB = XB$$

(proved earlier)

$$\angle PAB = \angle XCB$$

(proved earlier)

$$\angle BPA = \angle BXC = 120^\circ$$

(proved earlier)

$$\therefore \triangle PAB \equiv \triangle BXC$$

(AAS)

$$\therefore AP = XC$$

(matching sides of congruent $\triangle$ s)

$$\text{Now } PC = PX + XC$$

$$= PB + XC$$

($PX = PB$, sides of equilateral $\triangle PBX$)

$$= PB + AP$$

($AP = XC$, proved above)